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Community Detection on Signed Graphs: A Bayesian MCMC Coupling Approach

Seminar of the MathNet team (Inria Paris)

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Inria Paris, Room Paul Erdös (A115) – 2 p.m. Monday 15 June 2026

Outline

01
Motivation & the Sankararaman–Baccelli Model

Haplotype Assembly

1. Biological Setup: Two Homologous Chromosomes (Haplotypes)

Humans are diploid: they have two copies of each chromosome (homologous chromosomes).

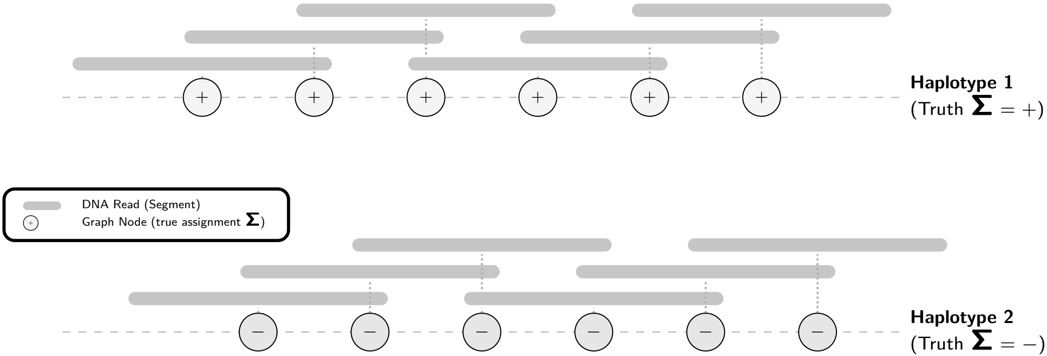
Haplotype 1
(Truth $\boldsymbol{\Sigma} = +$)

Haplotype 2
(Truth $\boldsymbol{\Sigma} = -$)

Haplotype Assembly

2. High-Throughput Sequencing: DNA Reads

Sequencing generates short, overlapping fragments called **reads** from both chromosomes.

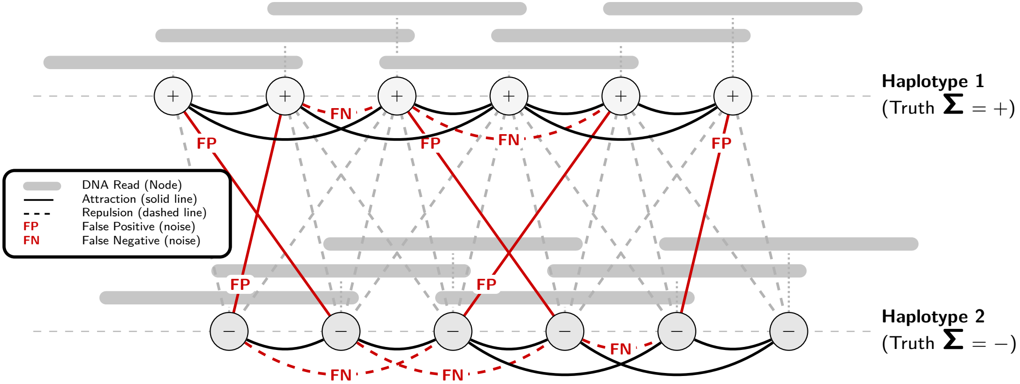


Haplotype Assembly

3. Modeling as a Signed Graph

Nodes = reads   Edges = overlaps

Goal: Reconstruct the true haplotype assignments $\boldsymbol{\Sigma}$ from this observed signed graph.



Motivation: Signed Community Detection

Community Detection on Signed Graphs ($K = 2$)

- We aim to partition nodes into $K = 2$ communities: $\mathbf{\Sigma}_i \in \{+1, -1\}$.
- Observed links represent **attractions** (solid lines) and **repulsions** (dashed lines).
- Under the hood, this is a **Stochastic Block Model** (SBM) setup.

The Geometric Challenge

- Nodes are embedded in space (e.g., genomic positions on $\mathbb{Z}$ or $\mathbb{R}^d$).
- Edge probabilities depend directly on the distance $r = \|x_i - x_j\|$:

$$\mathbb{P}(\text{attractive edge} \mid \text{same community}) = f_{\mathrm{in}}(r)$$
$$\mathbb{P}(\text{repulsive edge} \mid \text{different communities}) = f_{\mathrm{out}}(r)$$

- **All the difficulty (and the interest!) comes from this geometric dependency.**

The Sankararaman–Baccelli Model (GSBM)

Model Definition [1]

- **Spatial Nodes** V_n: Hom. Poisson PP $\mathcal{X}$ in $\mathbb{R}^d$ [2] restrict. to a n-volume window.
- **Latent Communities**: Uniform assignments $\Sigma_i \in \{+1, -1\}$ ($K = 2$).
- **Geometric Overlaps**: Observed connection $e = \{i, j\} \in H$ with probability:

$$\mathbb{P}(e \in H \mid X, \Sigma) = \begin{cases} f_{\mathrm{in}}(r_e) & \text{if } \Sigma_i = \Sigma_j \text{ (attraction)}, \\ f_{\mathrm{out}}(r_e) & \text{if } \Sigma_i \neq \Sigma_j \text{ (repulsion)}, \end{cases}$$

with distance $r_e = \|X_i - X_j\|$ and connecting functions $f_{\mathrm{in}}(r) \geq f_{\mathrm{out}}(r)$.

Goal: Weak Recovery

- Find an algorithm τ (independent of Σ) reconstructing assignments:

$$\lim_{n \to \infty} \mathbb{E}[\mathrm{ov}_n(\Sigma, \tau)] \geq \frac{1}{2} + \eta \quad \text{for some } \eta > 0,$$

using the **Community Overlap**: $\mathrm{ov}_n(\Sigma, \tau) = \max_{\pi \in \mathfrak{S}_2} \frac{1}{n} \sum_{i=1}^{n} \mathbb{I}_{\{\tau_i = \pi(\Sigma_i)\}}$.

Gibbs Model on Signed Graphs

Attractive and Repulsive Edges

We partition the potential edges $E = F \dot{\cup} \tilde{F}$:

- F: **Attractive** (ferromagnetic) edges, with weight $W_e > 0$
- $\tilde{F}$: **Repulsive** (antiferromagnetic) edges, with weight $W_e < 0$

Signed Gibbs Measure [3]

The probability of a label configuration $\sigma \in \{+1, -1\}^{V_n}$ is $\mu(\sigma) = \frac{1}{Z} e^{-U(\sigma)}$, where the energy $U(\sigma)$ penalizes unsatisfied edges:

$$U(\sigma) = \sum_{e=\{i,j\} \in F} W_e \mathbb{I}_{\{\sigma_i \neq \sigma_j\}} + \sum_{e=\{i,j\} \in \tilde{F}} |W_e| \mathbb{I}_{\{\sigma_i = \sigma_j\}}$$

The magnitudes $|W_e|$ determine the strength of each constraint.

From GSBM to the Gibbs Posterior

Conditioning on the Point Cloud

- With fixed spatial positions X, all pairwise distances $r_e = \|X_i - X_j\|$ are known.
- Under a uniform prior on Σ, the **posterior** distribution given the observed graph H is:

$$\mathbb{P}(\Sigma = \sigma \mid H, X) \propto \mathbb{P}(H \mid X, \sigma)$$

Log-Likelihood as Signed Gibbs Energy

- Choosing weights $W_e(H)$ as below:

$$W_e(H) = \begin{cases} \log\left(\dfrac{f_{\mathrm{in}}(r_e)}{f_{\mathrm{out}}(r_e)}\right) > 0 & \text{if } e \in H \text{ (attractive } \in F), \\ -\log\left(\dfrac{1 - f_{\mathrm{out}}(r_e)}{1 - f_{\mathrm{in}}(r_e)}\right) < 0 & \text{if } e \notin H \text{ (repulsive } \in \tilde{F}). \end{cases}$$

- ... yields the exact correspondence:

$$\mathbb{P}(H \mid X, \sigma) \propto e^{-U(\sigma)}$$

From GSBM to the Gibbs Posterior

Conditioning on the Point Cloud

- With fixed spatial positions X, all pairwise distances $r_e = \|X_i - X_j\|$ are known.
- Under a uniform prior on Σ, the **posterior** distribution given the observed graph H is:

$$\boldsymbol{\mathbb{P}(\Sigma = \sigma \mid H, X) = \mu(\sigma)} \propto \mathbb{P}(H \mid X, \sigma)$$

Log-Likelihood as Signed Gibbs Energy

- Choosing weights $W_e(H)$ as below:

$$W_e(H) = \begin{cases} \log\left(\dfrac{f_{\mathrm{in}}(r_e)}{f_{\mathrm{out}}(r_e)}\right) > 0 & \text{if } e \in H \text{ (attractive } \in F\text{)}, \\ -\log\left(\dfrac{1 - f_{\mathrm{out}}(r_e)}{1 - f_{\mathrm{in}}(r_e)}\right) < 0 & \text{if } e \notin H \text{ (repulsive } \in \tilde{F}\text{)}. \end{cases}$$

- ... yields the exact correspondence:

$$\mathbb{P}(H \mid X, \sigma) \propto e^{-U(\sigma)}$$

The Percolation Obstruction

Necessary Condition for Weak Recovery [1]

- Fragmented graphs (small, isolated components) prevent global reconstruction.
- Global information flow requires the graph to **percolate** [4] (possess a giant component).

Theorem (Sankararaman & Baccelli [1])

Independent bond percolation with edge probability

$$p_e = f_{\mathrm{in}}(r_e) - f_{\mathrm{out}}(r_e)$$

yielding a giant component is a ***necessary condition*** *for weak recovery.*

- **Our goal**: Retrieve and extend this result using unified **Bayesian MCMC coupling**.

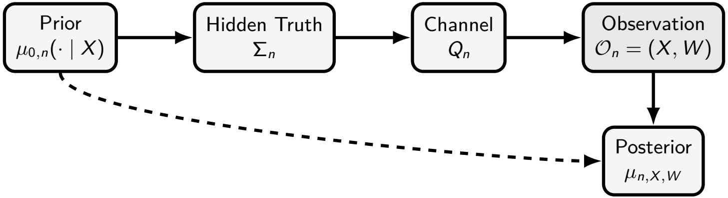
02
The Bayesian Framework & Coupling

The General Bayesian Model

Definition

1. **Nodes, Edges, Communities**: Nodes $V_n = \{1, \ldots, n\}$, edges E_n, communities $\mathcal{L}_K = \{0, \ldots, K-1\}$, and set of configurations $\sigma \in \mathcal{C}_n = (\mathcal{L}_K)^{V_n}$.
2. **Features**: $X_n \in \mathcal{X}_n$ according to distribution ν_n.
3. **Prior on configurations**: $\mu_{0,n}(\mathrm{d}\sigma \mid x)$ over $\mathcal{C}_n$.
4. **(Noisy) channel**: $Q_n(\mathrm{d}w \mid x, \sigma) = \prod_{e \in E_n} Q_{e,n}(\mathrm{d}w_e \mid x_i, x_j, \sigma_i, \sigma_j)$.

The joint law is $\mathbb{P}_n(\mathrm{d}x, \mathrm{d}\sigma, \mathrm{d}w) = \nu_n(\mathrm{d}x)\mu_{0,n}(\mathrm{d}\sigma \mid x)Q_n(\mathrm{d}w \mid x, \sigma)$.



Posterior Resampling (Nishimori Identity)

Replacing Truth by a Replica

- Comparing the algorithm's output with the hidden truth Σ_n is equivalent to comparing it with a posterior replica $\Sigma_n^{(1)}$.

Theorem (Posterior Resampling / Nishimori Identity [5])

Let τ_n be any community detection algorithm. Let $\Sigma_n^{(1)} \sim \mu_{n,X_n,W_n}$ be a replica drawn independently from the posterior. For any bounded test function Ψ_n:

$$\mathbb{E}\left[\Psi_n(X_n, W_n, \Sigma_n, \tau_n)\right] = \mathbb{E}\left[\Psi_n(X_n, W_n, \Sigma_n^{(1)}, \tau_n)\right].$$

Corollary (Overlap Identity)

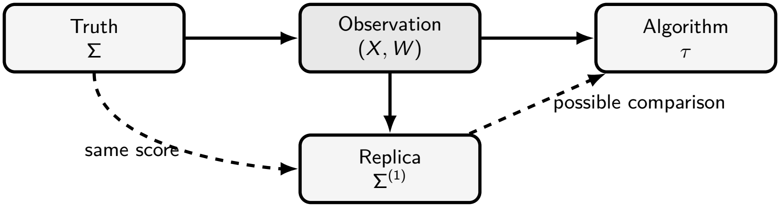
For any algorithm τ_n and threshold $s \in [0,1]$:

$$\mathbb{P}\left[\mathrm{ov}_n(\Sigma_n, \tau_n) \geq s\right] = \mathbb{P}\left[\mathrm{ov}_n(\Sigma_n^{(1)}, \tau_n) \geq s\right].$$

Invariant Coupling

Markov Chain Dynamics as a Source of Coupling

- We construct the posterior replica $\Sigma^{(1)}$ starting from the hidden truth Σ.
- Running **exactly one step** of a posterior-invariant $\textsc{Markov}$ kernel M initialized at Σ yields a valid replica $\Sigma^{(1)} \sim M(\Sigma, \cdot)$.

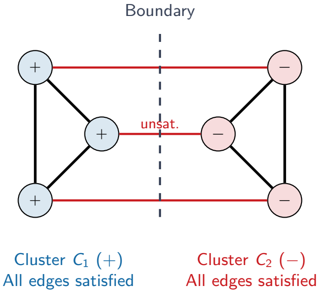


03
MCMC Dynamics & Performance Bounds

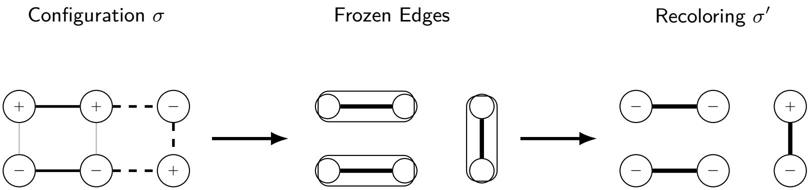
Single-Spin Dynamics & Geometric Bottleneck

Single-Spin Update Limitations

- **Geometry makes updates slow**: Single-spin dynamics (GLAUBER / METROPOLIS–HASTINGS [6]) update one node at a time.
- **Contradictory Clusters**: Two internally valid (highly satisfied) but mutually contradictory clusters are locked.
- **High Energy Barrier**: Flipping a single boundary node violates internal edges, creating a local energy barrier.
- **Conclusion**: Need for geometry $\Longrightarrow$ **Cluster Dynamics** (updating whole regions).



One Step of Signed Swendsen–Wang [7, 8]



- Edges in bold represent satisfied interactions that are randomly frozen.
- Clusters are then flipped independently.

Swendsen–Wang Signed Cluster Dynamics

Swendsen–Wang (SW) Signed Cluster Dynamics [7, 8]

- Update whole clusters of vertices simultaneously to speed up mixing.
- **Freezing Step**: For each edge $e = \{i, j\}$:
  - If e is satisfied ($W_e > 0$ and $\sigma_i = \sigma_j$, or $W_e < 0$ and $\sigma_i \neq \sigma_j$), freeze it with probability:

$$p_e^{\mathrm{frozen}} = 1 - e^{-|W_e|}.$$

  - If e is unsatisfied, do not freeze it ($p_e^{\mathrm{frozen}} = 0$).
- **Recoloring Step**: Flip each frozen cluster independently with probability $1/2$.

High-Order Triangular Dynamics

The Frustration Obstruction

- **Frustration**: $\nexists \sigma$ satisfying all edges (conflicting constraints).
- **Excessive Freezing**: In frustrated configurations, standard SW satisfies and freezes too many edges independently, blocking/freezing the MCMC dynamics.
- **Higher-Order Correlation**: Group interactions into **triangles** to correlate freezing and **freeze as little as possible**.

Triangular Cluster Dynamics [9]

- **Energy Decomposition**: Distribute edge energy U over triangles: $U(\sigma) = \sum_{t \in \mathcal{T}} U_t(\sigma)$.
- **Freezing Mechanism**: Apply a joint freezing mechanism on the triangles (rather than independent edges) to keep cluster sizes active.

Triangular Dynamics Freezing Rules

△ **Attractive Triangle** (Isotropic)					
Configuration	$\varnothing$	$\{ab\}$	$\{ac\}$	$\{bc\}$	$\{ab, ac, bc\}$
$(+,+,+)$	b_u	0	0	0	a_u
Other	1	0	0	0	0

Repulsive Triangle (Isotropic)					
Configuration	$\varnothing$	$\{ab\}$	$\{ac\}$	$\{bc\}$	$\{ab, ac, bc\}$
$(+,+,+)$	1	0	0	0	0
$(+,+,-)$	b_u	0	$a_u/2$	$a_u/2$	0
$(+,-,+)$	b_u	$a_u/2$	0	$a_u/2$	0
$(+,-,-)$	b_u	$a_u/2$	$a_u/2$	0	0

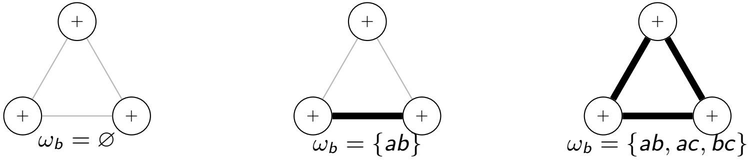
Table: Freezing rules for triangular dynamics ($a_u = 1 - e^{-2u}$, $b_u = e^{-2u}$), where u is the energy transferred by each edge.

General Swendsen–Wang Cluster Dynamics

General Framework [10]

Suppose the Hamiltonian decomposes as: $U(\sigma) = U_0(\sigma) + \sum_{b \in \mathcal{B}} U_b(\sigma)$, where each $b \in \mathcal{B}$ is a **bond** (e.g. edge, triangle, clique).

1. **Freezing Step**: For each bond $b \in \mathcal{B}$, freeze a sub-bond $\omega_b \subseteq b$ with probability $P_b^\sigma(\omega_b)$.
2. **Recoloring Step**: Recolor each frozen cluster independently according to a distribution preserving the prior $\mu_0(\sigma) \propto e^{-U_0(\sigma)}$.



General Swendsen–Wang Cluster Dynamics

Definition (Local Reversibility)

For two configurations σ, σ' and a bond b, we define the compatibility probability:

$$P_b^{\sigma\to\sigma'}(\omega_b) = \begin{cases} P_b^{\sigma}(\omega_b), & \text{if } P_b^{\sigma'}(\omega_b) > 0, \\ 0, & \text{otherwise.} \end{cases}$$

The freezing rule is **locally reversible** if for all b, ω_b, σ, σ':

$$P_b^{\sigma\to\sigma'}(\omega_b) = e^{-(U_b(\sigma')-U_b(\sigma))} P_b^{\sigma'\to\sigma}(\omega_b)$$

Theorem (Gibbs Invariance [10])

If freezing choices are independent, locally reversible, and $P_b^{\sigma}(\varnothing) > 0$ for all b, σ:

1. *The global cluster dynamics is reversible with respect to $\mu(\sigma) \propto e^{-U(\sigma)}$.*
2. *The condition $P_b^{\sigma}(\varnothing) > 0$ ensures the* ***irreducibility*** *of the Markov chain.*

Percolation as a Necessary Condition for Recovery

Theorem (Percolation Bound on Overlap [10])

Under the uniform prior μ_0, if frozen clusters are recolored independently and uniformly, then for the one-step posterior replica $\Sigma_n^{(1)}$, any algorithm τ_n, and any $\eta > 0$:

$$\lim_{n\to\infty} \mathbb{P}\left[\mathrm{ov}_n(\Sigma_n^{(1)}, \tau_n) \geq \frac{1}{K} + \frac{K-1}{K}(\theta^{\max} + \eta)\right] = 0$$

where $\theta^{\max}$ is the asymptotic fraction of nodes in giant frozen components.

Corollary (Percolation is Necessary)

If there is no percolation ($\theta^{\max} = 0$), the maximum overlap is bounded by $1/K$ (random guess): weak recovery is impossible. Hence, ***percolation is a necessary condition for weak recovery****.*

Key Intuition

Small clusters are recolored independently, washing out their contribution. Only giant (macroscopic) components can carry global community information.

Coupling & the Sankararaman–Baccelli Bound

One Step of Classical Edge-by-Edge Swendsen–Wang

Let's run one step of standard SWENDSEN–WANG (edges) initialized at the hidden truth Σ. For any edge e, the probability that it is frozen is:

- **Case 1 ($\Sigma_i = \Sigma_j$)**:

$$\mathbb{P}(\text{frozen}) = f_{\text{in}}(r_e)\left(1 - e^{-|W_e|}\right) = f_{\text{in}}(r_e)\left(1 - \frac{f_{\text{out}}(r_e)}{f_{\text{in}}(r_e)}\right) = f_{\text{in}}(r_e) - f_{\text{out}}(r_e)$$

- **Case 2 ($\Sigma_i \neq \Sigma_j$)**:

$$\mathbb{P}(\text{frozen}) = (1 - f_{\text{out}}(r_e))\left(1 - e^{-|W_e|}\right) = (1 - f_{\text{out}}(r_e))\left(1 - \frac{1 - f_{\text{in}}(r_e)}{1 - f_{\text{out}}(r_e)}\right) = f_{\text{in}}(r_e) - f_{\text{out}}(r_e)$$

Corollary (Retrieving the S&B Bound [1])

Under the classical edge SW dynamics, the frozen graph is ***exactly*** *an independent bond percolation with edge probability:*

$$p_e = f_{\text{in}}(r_e) - f_{\text{out}}(r_e).$$

Strict Improvement of the S&B bound (on the triangular grid)

Homogeneous Model ($f_{\mathrm{in}} = p$, $f_{\mathrm{out}} = 1 - p$)

On the triangular grid, we select a family of independent white triangles (like a chessboard).

- **Edge Dynamics (Standard SW)**: Frozen graph is independent edge percolation with parameter $p_{\mathrm{edge}} = 2p - 1$. Critical threshold: $p_c^{\mathrm{edge}} = \frac{1}{2} + \sin(\frac{\pi}{18}) \approx 0.673\ldots$.
- **Triangular Dynamics**: Frozen graph yields correlated triangle percolation. Critical threshold: $p_c^{\triangle} = \frac{7}{4} - \frac{\sqrt{17}}{4} \approx 0.7192\ldots$.

Corollary (Strictly Better Impossibility Bound [9])

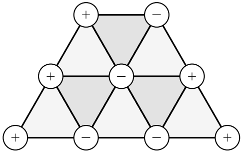
Since $p_c^{\mathrm{edge}} < p_c^{\triangle}$, for any

$$p \in [0.673\ldots, 0.7192\ldots),$$

standard edge dynamics percole (no conclusion), whereas triangular dynamics do not. Hence, we prove that weak recovery is impossible in this wider range of parameters!

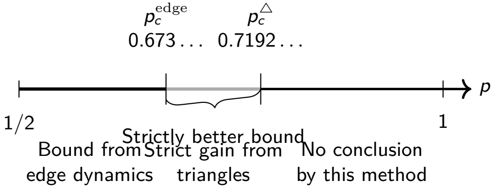
Comparison of Percolation Thresholds

Triangular Grid



White triangles used by triangular dynamics

Weak Recovery Impossibility Bounds



04
Open Questions & Conclusion

Open Questions

1. **Extension to Other Lattices and Higher Dimensions**
   - What are the thresholds on the Kagome, FCC, HCP, or Hyper-cubic lattices?
   - Can we define tetrahedral or higher-order simplex dynamics to obtain even tighter bounds?
2. **Multi-Community case ($K > 2$)**
   - How does the cluster dynamics extend when the spins take K values?
   - What is the role of Potts-like percolation in the coupling?
3. **Sufficiency of the Percolation Condition**
   - The percolation threshold provides a necessary condition (impossibility).
   - Is percolation also sufficient for the existence of a reconstruction algorithm?

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Merci.